

BRAHMI SARAH KAOUTHAR

CYBERSECURITY ENGINEERING STUDENT | AI SECURITY • THREAT INTELLIGENCE • SECURITY RESEARCH

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PROFILE

5th-year Cybersecurity Engineering student at ESI Sidi Bel Abbès, graduating in 2027. Focused on AI-driven security, threat intelligence, malware investigation, security detection, and adversarial cyber defense. Hands-on work spans endpoint telemetry, network security, reinforcement-learning attack/defense simulation, and applied ML.

RESEARCH FOCUS

AI Security · Threat Intelligence · Malware Analysis & Digital Forensics · Adversarial Security

EDUCATION

Cybersecurity Engineering — ESI Sidi Bel Abbès

2022–2027 · Expected graduation 2027

Baccalaureate in Mathematics — 16.41/20 · 2022

TECHNICAL SKILLS

Cybersecurity
Malware investigation · Digital forensics · Network security · Threat intelligence · Security detection
Security / SIEM
Wireshark · Nmap · Suricata · Sysmon · Winlogbeat · ELK · GNS3 · Volatility · Autopsy
AI / ML
Python · PyTorch · BERT · NLP · Deep learning · Reinforcement learning · OpenCV
Infrastructure
Docker · Ansible · Prometheus · Grafana · ZeroTier · LocalStack
Development
FastAPI · React.js · Flutter · MySQL · MongoDB · Cassandra · Neo4j · Firebase
Languages
Python · JavaScript · Dart · Java · C · Bash

LANGUAGES

Arabic (native) · French (fluent) · English (fluent)

ACTIVITIES

Alphabit Club (2023–present) · GDG Club (2022–2023)

EXPERIENCE

Network & Security Intern — Algérie Télécom

Complexe Haddadin, Oran · September 2025 · 3 weeks

Carrier-grade network and security exposure: MPLS, BGP, CGNAT, IPv6 transition strategies, Huawei OM monitoring, and critical-infrastructure security.

SELECTED PROJECTS

Cyber-Arena — Adversarial RL · Team Lead · 2026

- 18-node GNS3 enterprise environment for autonomous attack/defense research.
 - Deep Q-Network agents, SNMP telemetry, Telegraf, InfluxDB, Grafana, Ansible.
- WormLocker 2.0 — Ransomware Simulation & Detection · Team Project · 2025
- Ransomware behavior and detection through endpoint/network telemetry.
 - Sysmon, Winlogbeat, Elasticsearch/Kibana, Suricata, Volatility, Autopsy, MITRE ATT&CK.;

LTE-2G Downgrade Attack · Security Research · 2026

- Educational simulation of LTE-to-2G downgrade attacks, IMSI-catcher vulnerabilities, and SS7 weaknesses.

AI / ML Projects · 2025

- BERT semantic similarity for AISM; U-Net + EfficientNet-B3 for image segmentation.

Serverless Function Simulator · 2026

- Python/FastAPI FaaS runtime with HTTP, SQS, cron triggers, container pooling, and telemetry.

CTF & RESEARCH

CTF & Security Research — RSA lattice attacks (Coppersmith/CRT) and technical writeups.

Research Notes — AI for Cybersecurity / human-machine malware classification.